

Jim Geng

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Education

University of British Columbia

BSc. Computer Science & Statistics (Grade: 4.0/4.0)

Expected Dec 2027

Vancouver, BC

- **Relevant Coursework:** Software Engineering, Algorithm Design & Analysis, Operating Systems, Computer Networking, Statistical Modelling & Inference, Machine Learning

Experience

Teck Resources

Data Engineer Intern

May 2025 – Apr 2026

Calgary, AB

- Reduced annual compute costs by ~\$70K by consolidating 20 continuously running Databricks ingestion jobs into a single multi-threaded pipeline.
- Streamlined 10+ data pipeline request workflows by designing a configuration-driven form, then porting a dynamic React prototype to Microsoft PowerApps while preserving YAML-based form generation.
- Improved traceability of equipment data by building a custom Databricks workflow that reconciled asset management records using curated spreadsheets and source data, with SCD-style versioning to preserve history.
- Supported Teck's central data platform by developing and maintaining data pipelines on multiple architectures using Python, C#, and Scala, while automating infrastructure and CI/CD with Terraform, GitHub, and Azure DevOps.

Checker Software

Software Engineer Intern

Jan 2024 – Aug 2024

Vancouver, BC

- Built a print-ready PDF comparison export by exposing Pdfium WebAssembly page-import and drawing APIs to TypeScript, using native document handles and heap memory to compose and highlight PDFs side by side.
- Increased spreadsheet comparison capacity from ~10K to 1M rows by rebuilding the interface around DOM virtualization with variable-height content.
- Made spreadsheet exports ~5x faster in local testing by generating workbooks directly from a structured cell-diff model, enabling virtualization compatibility and consolidating two legacy dependencies into one.
- Developed full-stack on Diffchecker, a multi-format data comparison tool built with TypeScript, Next.js, Electron, Express, and PostgreSQL.

UBC Uncrewed Aircraft Systems

Software Team Member

Sept 2023 – Apr 2025

Vancouver, BC

- Led the design and development of a React-based control panel for autonomous flight planning, enabling custom UI views and workflows for UAV data visualization and competition-specific tasks.
- Contributed across multiple repositories by refactoring REST APIs and using Socket.io, Django, and SQLite to improve data processing and enable real-time telemetry workflows.

Projects

WiserSplit | TypeScript, React, Cloudflare Workers, D1, Hono, Drizzle ORM, Better Auth, Vitest

- Built a serverless expense-sharing application with React, Cloudflare Workers, Hono, D1, and Drizzle, supporting groups, flexible expense splits, settlements, and balance tracking.
- Implemented financial calculations using integer minor units and deterministic rounding to maintain consistent expense totals across participants.
- Added Google authentication, group-based access controls, automated tests, and persistent activity history for expense and settlement workflows.

Technical Skills

Languages: TypeScript, Python, Java, C#, C++, C, Golang, SQL

Technologies: React, Redux, Git, Node.js, Next.js, Express.js, Django, .NET, MSSQL, Databricks, Apache Spark, GitHub, Azure DevOps, PostgreSQL, Terraform, Linux, Docker, Tailwind CSS, LaTeX, Figma, Adobe Creative Suite

Concepts: Software Engineering, Machine Learning, Database Normalization, Agile, Cloud Computing